



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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**TECHNICAL
PRESENTATION**

Porter Stemmer: Design, Working Mechanism, and Applications

Presented by:

Name: G.VINAY KUMAR REDDY

Reg. No: 192424245

Course Code: DSA03

Course: Natural Language Processing

What Is Stemming?

Reducing words to their root/base form for text processing

1

The Problem

Words like "run", "running", "runs", and "ran" all mean the same core idea, but computers see them as completely different strings.

2

The Fix: Stemming

Stemming strips suffixes (like -ing, -ed, -s) to reduce related words down to one common "stem", so a search engine or classifier can treat them as one.

3

Porter's Solution (1980)

Martin Porter designed a fast, rule-based algorithm that applies a fixed sequence of suffix-stripping rules, without needing a dictionary.

How the Algorithm Works

1

Step 1

Plurals & -ed/-ing

caresses → caress

2

Step 2

Derivational suffixes

relational → relate

3

Step 3

Further suffix trims

electriciti →
electric

4

Step 4

Remove -ic, -ful, -ness

hopeful → hope

5

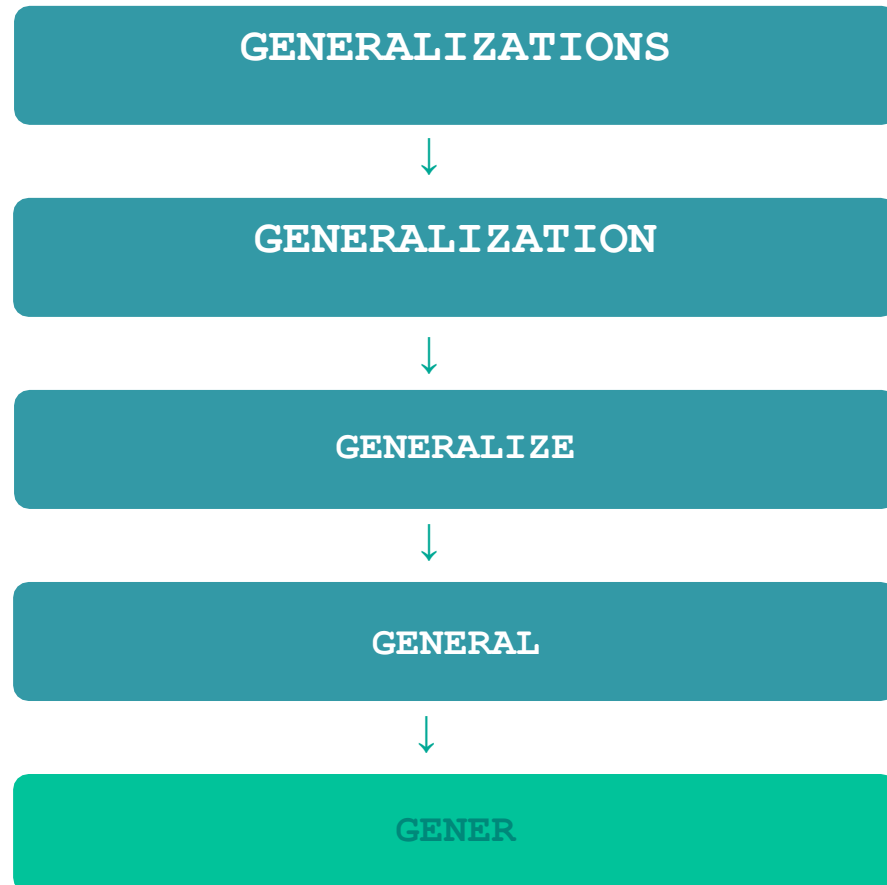
Step 5

Clean trailing -e / -l

controll → control

Example Walkthrough

Stemming the word “GENERALIZATIONS” into Key-terms



Final Stem

gener

All of “general,” “generalize,” and “generalization” now collapse to the same searchable token — exactly what a search engine index needs.



SAMPLE CODE AND OUTPUT

```
▶ from nltk.stem import PorterStemmer
ps = PorterStemmer()
words = [
    "playing",
    "played",
    "plays",
    "running",
    "runner",
    "studies",
    "studying",
    "happiness",
    "connected",
    "connection"
]

print("Original Word\tStemmed Word")
print("-" * 30)
for word in words:
    print(f"{word}\t\t{ps.stem(word)}")
```

```
... Original Word  Stemmed Word
-----
playing          play
played           play
plays            play
running          run
runner           runner
studies          studi
studying         studi
happiness        happi
connected        connect
connection       connect
```

Applications



Where It's Used

- Search engines — matching “running” to pages about “run”
- Information retrieval & document indexing
- Text classification and spam filtering
- Preprocessing for topic modeling (e.g. LDA)



Engineer to Excel



Limitations

- Over-stemming: “university” → “univ”, wrongly matching “universe”
- Under-stemming: doesn't always fully unify related words
- Produces non-words (e.g. “gener”), not real dictionary forms
- English-specific rules — doesn't generalize to other languages